

The herbal medicine *Dipsacus asper* Wall extract reduces the cognitive deficits and overexpression of β -amyloid protein induced by aluminum exposure

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Abstract

Excess aluminum (Al) exposure impairs neurocognitive function in humans and animals. Epidemiologic studies have shown a potential link between chronic Al exposure and Alzheimer's disease. In the present study, we sought to evaluate the protective effects of the herbal medicine *Dipsacus asper* extract against the cognitive impairment and overexpression of hippocampal β -amyloid protein (A β) induced by chronic Al exposure in rats. Vitamin E (VE) was used as a positive control. Following exposure to 0.3% aluminum chloride (AlCl₃) solution for 90 days in their drinking water, animals displayed a striking decrease (>80%) in step-through latency in the passive avoidance task and a significant increase (123%) in the number of A β immunoreactive cells in the hippocampus compared to controls. Al-exposed animals were then randomly assigned to receive vehicle, *Dipsacus asper* extract (4 g/kg), or VE (40 mg/kg) treatment up to 5 months. Both *Dipsacus asper* extract and VE significantly ameliorated animal's performance impairment in the passive avoidance task and suppressed the overexpression of hippocampal A β immunoreactivity. The effects of *Dipsacus asper* extract, but not VE, increased with time of treatment. The present results suggest that *Dipsacus asper* extract may possess therapeutic effects against Alzheimer's disease.

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Keywords: Aluminum; *Dipsacus asper*; Vitamin E; Alzheimer's disease; Passive avoidance task; β -amyloid protein; Rat

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Introduction

Aluminum (Al) is one of the most common metals existing in our environment. In the past three decades, Al in the etiology and pathogenesis of Alzheimer's disease has drawn growing attention, because epidemiologic studies have shown a potential link between chronic Al exposure and the illness (see review: [Flaten, 2001](#)). Al-induced cognitive deficits have been well documented from clinical observations ([Akila et al., 1999](#); [van Rensburg et al., 2001](#)) and animal experiments ([Alleva et al., 1998](#); [Colomina et al., 1999](#); [Golub et al., 1995](#)). Al may also be involved in other neurodegenerative disorders, such as amyotrophic lateral sclerosis and Parkinsonism-dementia of Guam ([Wills and Savory, 1983](#)) and dialysis encephalopathy ([Perl et al., 1982](#)).

Hallmark neuropathological features of Alzheimer's disease are neuronal cell loss with neurofibrillary tangles and senile plaques with amyloid deposits, in which β -amyloid protein ($A\beta$), derived from proteolytic cleavage of amyloid precursor protein, is one of the major components ([Hardy and Higgins, 1992](#)). A substantial number of studies have shown Al-induced neurofibrillary lesions and the formation of tangle-like structures in neuronal cells (see reviews: [Shin, 1997](#)). Recent studies have found that Al exposure altered conformation of $A\beta$ and enhanced its aggregation on the surface of cultured neurons ([Bondy and Truong, 1999](#); [Campbell et al., 2000](#); [Kawahara et al., 2001](#)).

Although the mechanisms of Al neurotoxic actions are not fully understood, multiple pathways are likely to be involved. One of them, Al-induced brain lipid peroxidation may be a key component of Al neurotoxicity ([Ohyashiki et al., 1998](#); [Swain and Chainy, 1997](#); [Verstraeten et al., 1998](#); [Verstraeten and Oteiza, 2000](#)). Vitamin E (VE, α -tocopherol), a powerful biological antioxidant, is effective in improving the cognitive performance of aged rats ([Socci et al., 1995](#)), preventing brain injury, and inhibiting senile plaque deposits induced by Al-induced oxidative stress ([Abd el-Fattah et al., 1998](#); [Pratico et al., 2002](#)).

The herbal medicine *Dipsacus asper* Wall (Dipsacaceae) has long been used as a tonic and anti-inflammatory agent in traditional Chinese medicine for the therapy of low back pain, knee pain, rheumatic arthritis, traumatic hematoma, threaten abortion, and bone fractures ([Wang, 1999](#); [Zhou et al., 1981](#)). Several chemical constituents, particularly *Dipsacus* saponins, have been identified from the root of *Dipsacus asper* ([Jung et al., 1993](#); [Zhang and Xue, 1991a,b, 1992, 1993](#)). *Dipsacus* saponins have been demonstrated to possess antinociceptive ([Suh et al., 1996, 2000](#)) and anticomplementary effects ([Oh et al., 1999](#)). *Dipsacus asper* was also reported to be effective in treating VE-deficiency ([Xie et al., 1994](#)).

We report here that chronic treatment with *Dipsacus asper* extract reduced the cognitive deficits in the passive avoidance task and the overexpression of hippocampal $A\beta$ immunoreactivity in rats chronically exposed to Al, using VE as a positive control.

Materials and methods

Animals

A total of 99 male Sprague-Dawley rats (~ 200 g), provided by Experimental Animal Center, Xi'an, Shaanxi, were used in this study. Animals were housed stainless-steel cages in a room maintained in a 12-h light/dark cycle at around 22 °C. Food and water were available ad libitum.

Animals were allowed to acclimate for at least 10 days prior to the experiment. All animal care and treatment were conducted in accordance with the regulations of Xi'an Jiaotong University for the use of experimental animals in research, and in conformity with the NIH Guide for the Care and Use of Laboratory Animals.

Preparation of drugs

The dried root of *Dipsacus asper*, originally grown in Sichuan Province of China and harvested in a July–August period, was obtained from a local Chinese herbal store in Xi'an, Shaanxi. Its identity was authenticated by an herbalist, and a sample specimen is kept in Department of Human Anatomy, College of Medicine, Xi'an Jiaotong University, Xi'an, Shaanxi, China. To minimize the variation in composition of active ingredients, the herbal materials were ensured to come from the same habitat.

The extractive procedure of *Dipsacus asper* has been described previously (Qian et al., 1999). Briefly, the dried root of *Dipsacus asper* (16 kg) was roughly grounded and refluxed with EtOH twice (1 h for each extraction). The ethanol extracts were filtered and lyophilized to obtain the powder *Dipsacus asper* extract, in which the most abundant bioactive component *Dipsacus* saponins has been well characterized previously (Jung et al., 1993; Wang, 1999; Zhang and Xue, 1993).

For oral administration, the extractive powder of *Dipsacus asper* and the positive control VE were dissolved in the vehicle vegetable oil at a concentration of equivalent 4 gram raw drugs per ml and 40 mg/ml, respectively.

Passive avoidance task

The apparatus consisted of an illuminated chamber (20 × 25 × 30 cm) and a dark chamber (40 × 25 × 30 cm) with a metal grid floor. The two chambers were separated by a wall with a guillotine door (8 cm in diameter). The test was performed on 2 consecutive days. In the acquisition trial, rats were individually placed in the illuminated chamber. Once entering the dark chamber, an electric shock (40 V, 0.5 mA, 1 s) was delivered to the feet through the floor grid. The rat was immediately removed from the apparatus and returned to its home cage. At the retention trial, conducted 24 h later, the rat was placed again in the illuminated chamber, and the interval (step-through latency) between the placement in the illuminated chamber and the entry into the dark chamber was recorded. If the animal did not enter the dark chamber within a 5-min test period, the test was terminated and the step-through latency was recorded as 300 s.

Each rat was tested at the end of Al exposure, and the end of 1, 3 and/or 5 months of drug treatment. Acquisition and retention trials were included for each test because the avoidance response disappears 26 days after acquisition trial (Ichitani et al., 1992).

Experimental design

Normal controls (n = 15) and Al-exposed rats (n = 84) received distilled water and 0.3% aluminum chloride (AlCl₃) solution in distilled water, respectively, as drinking water from the cage tap for 90 days. Despite a suppression of rats' thirst desire was observed, which might be due to changes in pH values of the drinking water by AlCl₃ solution, the rate of water consumption seemed to be not affected significantly.

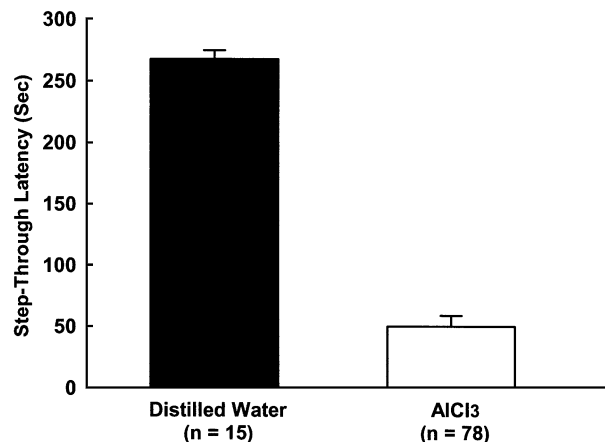


Fig. 1. The effects of chronic Al exposure on cognitive function of rats. Rats received distilled water or 0.3% AlCl₃ solution as drinking water from the cage tap for 90 days, and then tested in the passive avoidance task. The step-through latency was recorded from the retention trial. Data are expressed as mean $\pm$ S.E.M. The latency of Al-exposed rats is only 19% of the control value.

All rats were tested in the passive avoidance task. Al-exposed rats whose step-through latency was decreased over 60% compared to the average value of controls (78 out of 84) were selected, and 6 of them were used for A β immunohistochemistry (see below). Other 72 were randomly administered by oral gavage with vehicle, *Dipsacus asper* extract (4 g/kg) or VE (40 mg/kg). At the end of 1, 3 and 5 months of treatment, the rats were tested in the avoidance task 1 hour after administered vehicle or drugs, and then sacrificed in a batch (n = 8) for A β immunohistochemistry.

Normal controls did not receive drug treatment, but were tested in the passive avoidance task and sacrificed in a batch (n = 5) for A β immunohistochemistry in the same manner.

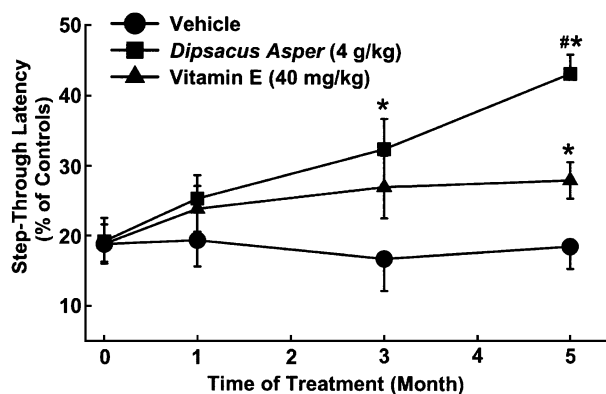


Fig. 2. The reversal effects of long-term treatment with *Dipsacus asper* extract and VE against the decreased step-through latency of Al-exposed rats. The rats were given orally vehicle, *Dipsacus asper* extract (4 g/kg) or VE (40 mg/kg) up to 5 months, and tested in the passive avoidance task at the end of 1, 3 and 5 months of treatment. Data are expressed as percentage of matched normal controls with mean $\pm$ S.E.M. * P < 0.05 vs matched vehicle group, # P < 0.05 vs matched VE group (n = 24 at 0- and 1-month time point, n=16 at 3-month time point, and n=8 at 5-month time point).

A β immunohistochemistry

Animals were deeply anesthetized with 20% urethane and perfused transcardially with cold phosphate-buffered saline (PBS, 0.1 M, pH 7.4), followed by 4% paraformaldehyde. Brains were removed and soaked in 30% sucrose at 4 °C overnight. Coronal sections with the thickness of 40 μ m were cut on a vibrant microtome, and collected in Tris-saline (pH 7.6).

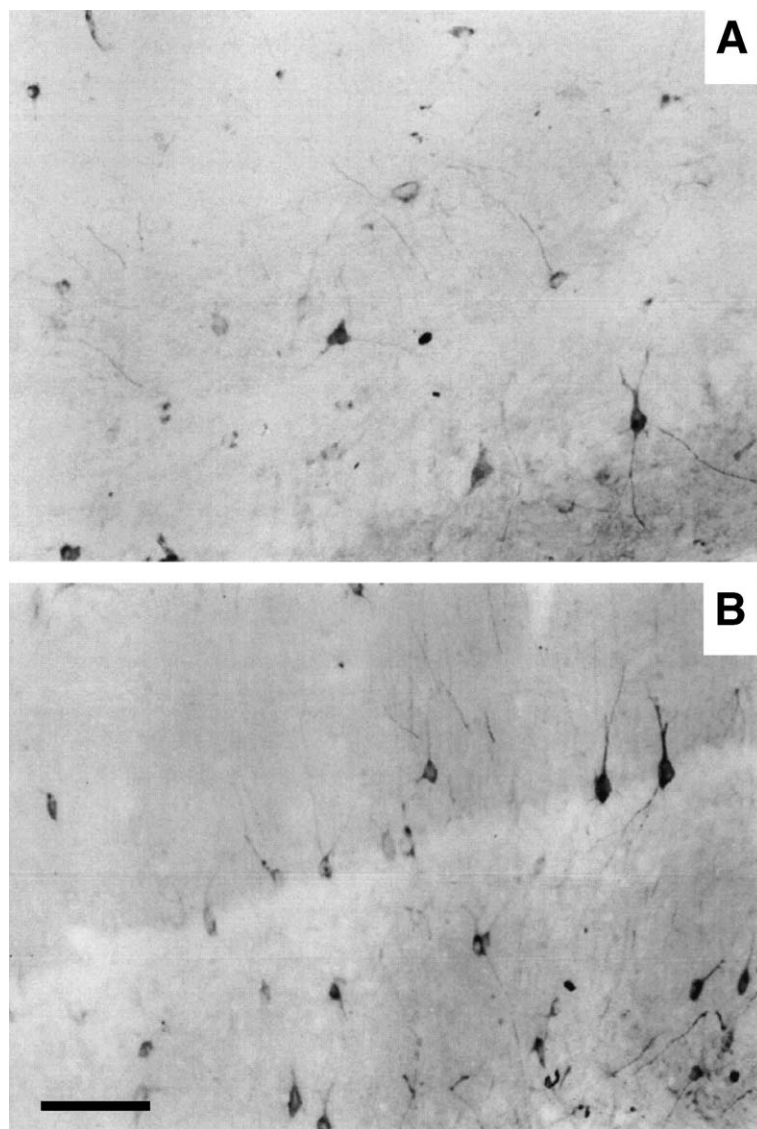


Fig. 3. Chronic Al exposure induced the overexpression of A β immunoreactivity in CA1 sector of the hippocampus. Rats received the drinking of distilled water (A) or 0.3% AlCl₃ solution (B) from the cage tap for 90 days, and their brains were then removed for A β immunohistochemistry. Al-exposed rats showed a striking increase in the number of A β immunoreactive cells compared to normal controls (Bar = 40 μ m).

Immunohistochemistry using indirect immunoperoxidase technique for A β was performed as previously described (Qian et al., 1997; Hu et al., 1997). Briefly, sections were pretreated with 90% formic acid for 5 min and 0.3% H₂O₂ for 20 min at room temperature, followed by 0.3% Triton X-10 and normal horse serum (1:50) for 30 min at 37 °C, respectively. Sections were then incubated in primary monoclonal mouse anti-A β raised against a synthetic peptide consisting of residues 8–17 of A β (1:1,000, clone 6F/3D, DAKO, Denmark) (Allsop et al., 1986) for 48 h at 4 °C. In control experiments, the primary antibody solution was excluded. All sections were then incubated in 1:200 biotinylated secondary goat antibodies raised against mouse and 1:50 avidin-biotin complex (ABC: Vector Laboratories, Burlingame, CA, USA) for 1 h each at 37 °C, successively. The sections were developed with 0.1% diaminobenzidine (DAB) and 0.004% H₂O₂ for 20 min at room temperature. To identify brain regions, some sections were counterstained with cresyl violet. During the above procedures, the sections were washed 3 to 4 times for 5 min each in cold PBS after completion of each step. The sections were finally dehydrated in increasing concentration of ethanol, cleared in xylene, and mounted under coverslips in DPX.

To minimize systematic variations, all staining procedures were conducted in batches containing equal proportions of brain sections from each group, and sections within a batch were processed under identical experimental conditions.

A β positive cell counts

Five matched sections from each rat were selected for cell counts. A β positive neurons in the hippocampus between Bregma –2.8 and –4.3 mm (Paxinos and Watson, 1986) were counted bilaterally at a magnification of 100 $\times$ under a light microscope. Only those exhibiting brown cytoplasmic A β immunostaining with sharply defined perimeter and nucleus were counted. The number of positively stained neurons was expressed as number of neurons per section.

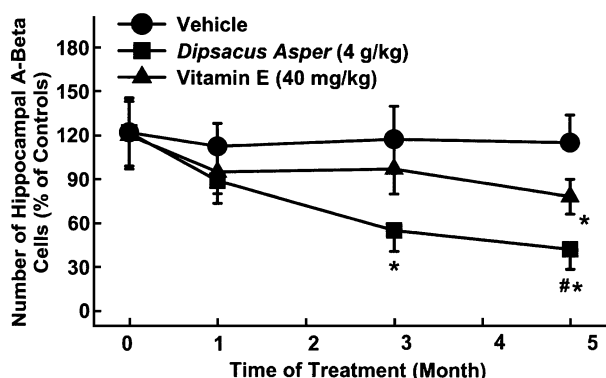


Fig. 4. The reversal effects of long-term treatment with *Dipsacus asper* extract and VE against the overexpression of A β immunoreactivity in the hippocampus of Al-exposed rats. The rats were given orally vehicle, *Dipsacus asper* extract (4 g/kg) or VE (40 mg/kg) up to 5 months, and sacrificed in a batch at the end of 1, 3 and 5 months of treatment. Brain sections were prepared for A β immunohistochemistry. Data are expressed as percentage of matched normal controls with mean $\pm$ S.E.M. * P < 0.05 vs matched vehicle group, # P < 0.05 vs matched VE group (n = 24 at 0- and 1-month point, n = 16 at 3-month point, and n = 8 at 5-month point).

Statistical analysis

The step-through latency and the number of A β positive neurons in vehicle- and drug-treated groups were transformed to percentage of matched normal controls. All data were expressed as mean $\pm$ S.E.M. Two-way analysis of variance (ANOVA) was used to evaluate group $\times$ duration of treatment

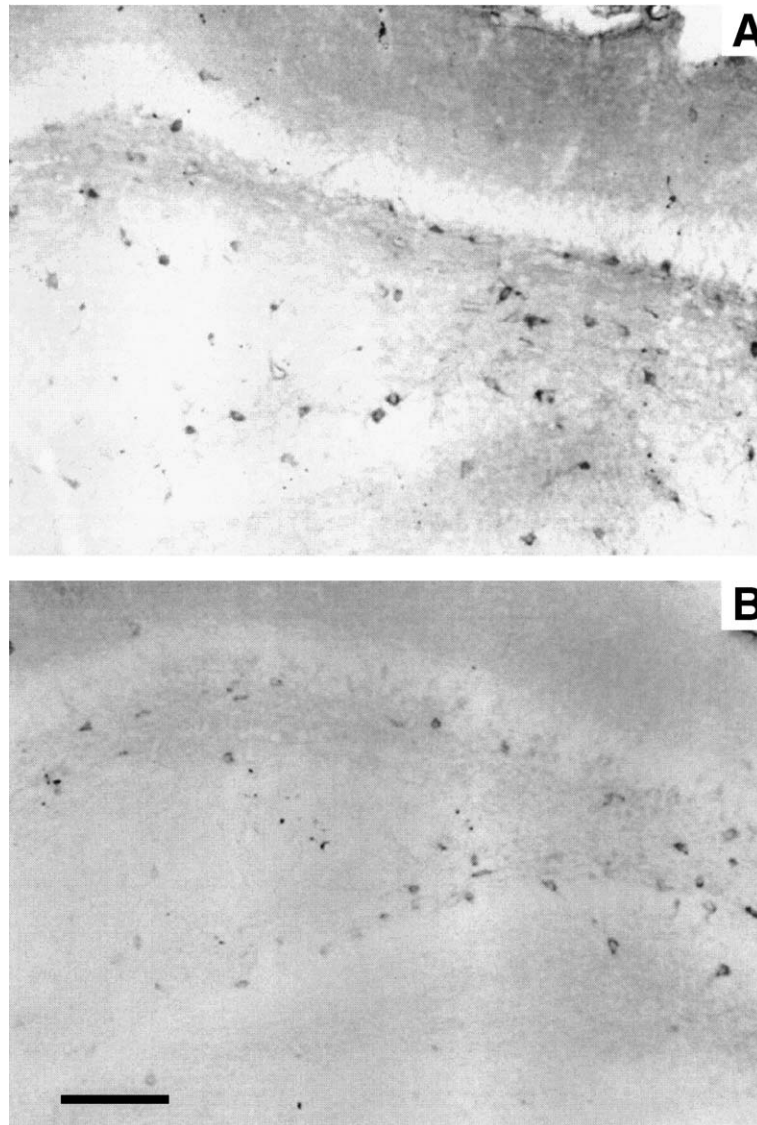


Fig. 5. The inhibitory effects of long-term treatment with *Dipsacus asper* extract on the overexpression of A β immunoreactivity in the hippocampus of AI-exposed rats. The rats were given orally vehicle (**A**) or *Dipsacus asper* extract (4 g/kg) (**B**) for 5 months, and then brains were removed for A β immunohistochemistry. *Dipsacus asper* extract treatment significantly decreased the number of A β positive cells (Bar = 100 μ m).

interaction, followed by post hoc multiple comparisons using Student-Newman-Keuls test to further analyze differences between groups. Statistical significance was defined as $p \leq 0.05$.

Results

Passive avoidance task

Seventy eight out of 84 AI-exposed rats showed a >60% decrease in the step-through latency, and their average value was 50.2 ± 8.8 s, which was only 19% of the control value (267.8 ± 7.2 s) (Fig. 1).

Fig. 2 shows the time course of the ability of *Dipsacus asper* extract and VE to decrease the step-through latency resulting from AI exposure. Two-way ANOVA analysis revealed a significant interaction between group $\times$ duration of treatment on the step-through latency ($F_{(2,3)} = 11.23$, $P < 0.05$). Multiple comparisons further showed that, at 3 and 5 month points, the latency of AI-exposed rats treated with *Dipsacus asper* was significantly extended compared to matched vehicle controls ($P < 0.05$). Furthermore, the extended latency of *Dipsacus asper* group was markedly greater than that of VE-treated group at the 5-month time point ($P < 0.05$) (Fig. 2).

A β immunohistochemistry

In normal controls, only scattered and faintly stained A β positive cells were present in the hippocampus (Fig. 3A). In AI-exposed rats, the distribution pattern of cell bodies immunolabeled for A β in the hippocampus was identical to that of normal controls, but the number of A β positive cells was increased by 123% (78 ± 7 per section) compared to the average value of normal controls (35 ± 7 per section). Most of these positive cells were localized in the pyramidal cell layer of the CA₁ and CA₃ sector and polymorphical cell layer of the dentate gyrus (Fig. 3B).

A significant interaction on the number of hippocampal A β positive cells was observed between group $\times$ duration of treatment ($F_{(2,3)} = 10.56$, $P < 0.05$). Multiple comparisons further revealed that a significant decrease in the number of the A β positive cells was evident in *Dipsacus asper*-treated rats at the end of 3 and 5 months (Figs. 4, 5), and VE-treated rats at the end of 5 months compared to matched vehicle-treated rats ($P < 0.05$). Moreover, the number of the A β positive cells of *Dipsacus asper*-treated rats was significantly decreased compared to that of VE-treated rats at the end of 5 months ($P < 0.05$).

Discussion

The principal findings of the present study are that long-term treatment with *Dipsacus asper* extract reduces the cognitive impairment and overexpression of hippocampal A β immunoreactivity induced by AI exposure. The effects of *Dipsacus asper* extract increase with time of treatment and exceed those of the antioxidant agent VE after 5 months of treatment.

In the present study, chronic exposure to AI reduces the avoidance response and increases the expression of A β immunoreactivity in the hippocampus. These results, merging with previous studies (Alleva et al., 1998; Campbell et al., 2000; Colomina et al., 1999; Golub et al., 1995; Kawahara et al.,

2001; Neill et al., 1996; Pratico et al., 2002), have provided direct evidence to support viewpoint that Al may be a potential contributing factor in the formation of neurofibrillary tangles and cognitive deficits in Alzheimer's disease. The present study found further that the cognitive deficits induced by Al exposure are ameliorated by chronic VE treatment. VE also suppresses the overexpression of hippocampal A β immunoreactivity in Al-exposed rats. Since VE has the ability to reduce oxidative stress and inhibit amyloid deposits (Butterfield et al., 1999; Miranda et al., 2000), the present results suggest that Al-induced cognitive and neurodegenerative changes are directly related to its capacity to cause oxidative brain injury and induce amyloid deposits.

In addition, considerable evidence has demonstrated that chronic Al exposure causes the dysfunction of several neurotransmitter enzymes, including choline acetyl transferase (ChAT), acetylcholinesterase (AChE) (Bielarczyk et al., 1998; Platt et al., 2001; Szutowicz, 2001), monoamine oxidase (MAO) (Zatta et al., 1998), and glutamate α -decarboxylase (GAD) (Nayak and Chatterjee, 2001). Al exposure also alters the concentrations of amino acid neurotransmitters, their metabolites, and the neuropeptides cholecystokinin (CCK) and substance P activity in several brain regions (Beal et al., 1989; Jia et al., 2001; Tsunoda and Sharma, 1999). Intraventricular microinjection of Al was reported to produce the inflammatory response of neuroglia (Platt et al., 2001). Thus, the neurotoxicity of Al is likely to be the result of a combination of several mechanisms, including oxidative brain injury, induced amyloid deposits and neuroglia inflammatory reaction, and reduced neurotransmitter biosynthesis.

Based on chemical constituents of *Dipsacus asper*, there are three possible mechanisms by which *Dipsacus asper* extract protects against the neurotoxic effects of Al. Firstly, the dried root of *Dipsacus asper* may contain VE and was reported to be effective in treating VE deficiency, although the abundance of VE is unknown (Wang, 1999; Zhou et al., 1981). It should be therefore assumed that, like VE, *Dipsacus asper* extract may have antioxidant effects, that may in part contribute to its suppression of A β immunoreactivity of Al-exposed rats. Whether the neuroprotective potency of *Dipsacus asper* extract correlates with the concentrations of VE contained in the root deserves to be further investigated. Secondly, *Dipsacus* saponins have been identified to be the most abundant bioactive constituent in the root of *Dipsacus asper* (Jung et al., 1993; Wang, 1999; Zhang and Xue, 1993). Among them *Dipsacus* saponin C has been found to possess the antinociceptive properties, which is mediated by multiple receptor systems, including GABA, glutamate, noradrenaline, serotonin and neuropeptides (Suh et al., 1996, 2000), suggesting that *Dipsacus* saponin C may have the capacity of inhibiting the interfering effects of Al on neurotransmitter biosynthesis. Finally, despite little is known about the mechanisms underlying the anti-inflammatory actions of *Dipsacus asper*, clinical observations have shown its effectiveness in treating rheumatic illnesses (Xie et al., 1994). *Dipsacus asper* extract may have similar effects against the Al-induced inflammatory response of neuroglia cells.

In summary, the present study demonstrates that long-term treatment with the herbal medicine *Dipsacus asper* extract significantly ameliorates the cognitive impairment in the passive avoidance task in rats chronically exposed to Al. *Dipsacus asper* extract also suppresses the Al-induced overexpression of hippocampal A β immunoreactivity. The protective effects of *Dipsacus asper* extract against the neurotoxicity Al are significantly greater than those of the antioxidant agent VE, suggesting that the protective properties of *Dipsacus asper* extract may derive from multiple mechanisms, including as an antioxidant. The potential relevance of the present results to clinical therapeutics in the treatment of Alzheimer's disease deserves further investigation.

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References

- Abd el-Fattah, A.A., al-Yousef, H.M., al-Bekairi, A.M., al-Sawaf, H.A., 1998. Vitamin E protects the brain against oxidative injury stimulated by excessive aluminum intake. *Biochemical and Molecular Biology International* 46, 1175–1180.
- Akila, R., Stollery, B.T., Riihimaki, V., 1999. Decrements in cognitive performance in metal inert gas welders exposed to aluminium. *Occupational and Environmental Medicine* 56, 632–639.
- Alleva, E., Rankin, J., Santucci, D., 1998. Neurobehavioral alteration in rodents following developmental exposure to aluminum. *Toxicology and Industrial Health* 14, 209–221.
- Allsop, D., Landon, M., Kidd, M., Lowe, J.S., Reynolds, G.P., Gardner, A., 1986. Monoclonal antibodies raised against a subsequence of senile plaque core protein react with plaque cores, plaque periphery and cerebrovascular amyloid in Alzheimer's disease. *Neuroscience Letters* 68, 252–256.
- Beal, M.F., Mazurek, M.F., Ellison, D.W., Kowall, N.W., Solomon, P.R., Pendlebury, W.W., 1989. Neurochemical characteristics of aluminum-induced neurofibrillary degeneration in rabbits. *Neuroscience* 29, 339–346.
- Bielarczyk, H., Tomaszewicz, M., Szutowicz, A., 1998. Effect of aluminum on acetyl-CoA and acetylcholine metabolism in nerve terminals. *Journal of Neurochemistry* 70, 1175–1181.
- Bondy, S.C., Truong, A., 1999. Potentiation of beta-folding of beta-amyloid peptide 25-35 by aluminum salts. *Neuroscience Letters* 267, 25–28.
- Butterfield, D.A., Koppal, T., Subramaniam, R., Yatin, S., 1999. Vitamin E as an antioxidant/free radical scavenger against amyloid beta-peptide-induced oxidative stress in neocortical synaptosomal membranes and hippocampal neurons in culture: insights into Alzheimer's disease. *Reviews of Neuroscience* 10, 141–149.
- Campbell, A., Kumar, A., La Rosa, F.G., Prasad, K.N., Bondy, S.C., 2000. Aluminum increases levels of beta-amyloid and ubiquitin in neuroblastoma but not in glioma cells. *Proceedings of the Society of Experimental Biology and Medicine* 223, 397–402.
- Colomina, M.T., Sanchez, D.J., Sanchez-Turet, M., Domingo, J.L., 1999. Behavioral effects of aluminum in mice: influence of restraint stress. *Neuropsychobiology* 40, 142–149.
- Flaten, T.P., 2001. Aluminium as a risk factor in Alzheimer's disease, with emphasis on drinking water. *Brain Research Bulletin* 55, 187–196.
- Golub, M.S., Han, B., Keen, C.L., Gershwin, M.E., Tarara, R.P., 1995. Behavioral performance of Swiss Webster mice exposed to excess dietary aluminum during development or during development and as adults. *Toxicology and Applied Pharmacology* 133, 64–72.
- Hardy, J.A., Higgins, G.A., 1992. Alzheimer's disease: the amyloid cascade hypothesis. *Science* 256, 184–185.
- Hu, H.T., Qian, Y.H., Yang, J., Zhang, Z.J., Wang, W.X., Fang, X.Y., 1997. Expression of amyloid precursor protein in dorsal hippocampal formation of dementia model in rat. *Chinese Journal of Anatomy* 20, 444–446.
- Ichitani, Y., Okaichi, H., Yoshikawa, T., Ibata, Y., 1992. Learning behaviour in chronic vitamin E-deficient and-supplemented rats: radial arm maze learning and passive avoidance response. *Behavioral Brain Research* 51, 157–164.
- Jia, Y., Zhong, C., Wang, Y., 2001. Effects of aluminum on amino acid neurotransmitters in hippocampus of rats. *Chinese Journal of Preventive Medicine* 35, 397–400.
- Jung, K.Y., Do, J.C., Son, K.H., 1993. Triterpene glycosides from the roots of *Dipsacus asper*. *Journal of Natural Products* 56, 1912–1916.
- Kawahara, M., Kato, M., Kuroda, Y., 2001. Effects of aluminum on the neurotoxicity of primary cultured neurons and on the aggregation of beta-amyloid protein. *Brain Research Bulletin* 55, 211–217.
- Miranda, S., Opazo, C., Larrondo, L.F., Munoz, F.J., Ruiz, F., Leighton, F., Inestrosa, N.C., 2000. The role of oxidative stress in the toxicity induced by amyloid beta-peptide in Alzheimer's disease. *Progress in Neurobiology* 62, 633–648.

- Nayak, P., Chatterjee, A.K., 2001. Effects of aluminium exposure on brain glutamate and GABA systems: an experimental study in rats. *Food and Chemical Toxicology* 39, 1285–1289.
- Neill, D., Leake, A., Hughes, D., Keith, A.B., Taylor, G.A., Allsop, D., Rima, B.K., Morris, C., Candy, J.M., Edwardson, J.A., 1996. Effect of aluminium on expression and processing of amyloid precursor protein. *Journal of Neuroscience Research* 46, 395–403.
- Oh, S.R., Jung, K.Y., Son, K.H., Park, S.H., Lee, I.S., Ahn, K.S., Lee, H.K., 1999. In vitro anticomplementary activity of hederagenin saponins isolated from roots of *Dipsacus asper*. *Archives of Pharmacological Research* 22, 317–319.
- Ohyashiki, T., Suzuki, S., Satoh, E., Uemori, Y., 1998. A marked stimulation of Fe²⁺-initiated lipid peroxidation in phospholipid liposomes by a lipophilic aluminum complex, aluminum acetylacetonate. *Biochemical and Biophysical Acta* 1389, 141–149.
- Paxinos, G., Watson, C., 1986. The rat brain in stereotaxic coordinates. Academic Press, New York.
- Perl, D.P., Gajdusek, D.C., Garruto, R.M., Yanagihara, R.T., Gibbs, C.J., 1982. Intraneuronal aluminum accumulation in amyotrophic lateral sclerosis and Parkinsonism-dementia of Guam. *Science* 217, 1053–1055.
- Platt, B., Fiddler, G., Riedel, G., Henderson, Z., 2001. Aluminium toxicity in the rat brain: histochemical and immunocytochemical evidence. *Brain Research Bulletin* 55, 257–267.
- Pratico, D., Uryu, K., Sung, S., Tang, S., Trojanowski, J.Q., Lee, V.M., 2002. Aluminum modulates brain amyloidosis through oxidative stress in APP transgenic mice. *FASEB Journal* 16, 1138–1140.
- Qian, Y.H., Hu, H.T., Yang, J., Zhang, Z.J., Wang, W.X., Yang, G.D., 1999. Effect of *Dipsacus asper* on the expression of amyloid precursor protein in the dorsal hippocampal formation in rat model of Alzheimer's disease. *Chinese Journal of Neuroscience* 15, 134–138.
- Qian, Y.H., Yang, J., Ren, H.M., Hu, H.T., Zhang, Z.J., 1997. Amyloid protein accumulation in dorsal hippocampal formation of a rat model of dementia: an immunocytochemical study. *Journal of Xi'an Medical University* 18, 304–307.
- Shin, R.W., 1997. Interaction of aluminum with paired helical filament tau is involved in neurofibrillary pathology of Alzheimer's disease. *Gerontology* 43 (S1), 16–23.
- Socci, D.J., Crandall, B.M., Arendash, G.W., 1995. Chronic antioxidant treatment improves the cognitive performance of aged rats. *Brain Research* 693, 88–94.
- Suh, H.W., Song, D.K., Huh, S.O., Son, K.H., Kim, Y.H., 2000. Antinoceptive mechanisms of *Dipsacus* saponin C administered intrathecally in mice. *Journal of Ethnopharmacology* 71, 211–218.
- Suh, H.W., Song, D.K., Son, K.H., Wie, M.B., Lee, K.H., Jung, K.Y., Do, J.C., Kim, Y.H., 1996. Antinoceptive mechanisms of *Dipsacus* saponin C administered intracerebroventricularly in the mouse. *General Pharmacology* 27, 1167–1172.
- Swain, C., Chainy, G.B., 1997. Aluminum effect on lipid peroxidation and on the activities of superoxide dismutase and catalase in the cerebral hemisphere and liver of young chicks. *Journal of Trace Element Medicine and Biology* 11, 77–82.
- Szutowicz, A., 2001. Aluminum, NO, and nerve growth factor neurotoxicity in cholinergic neurons. *Journal of Neuroscience Research* 66, 1009–1018.
- Tsunoda, M., Sharma, R.P., 1999. Altered dopamine turnover in murine hypothalamus after low-dose continuous oral administration of aluminum. *Journal of Trace Element Medicine and Biology* 13, 224–231.
- van Rensburg, S.J., Potocnik, F.C., Kiss, T., Hugo, F., van Zijl, P., Mansvelt, E., Carstens, M.E., Theodorou, P., Hurly, P.R., Emsley, R.A., Taljaard, J.J., 2001. Serum concentrations of some metals and steroids in patients with chronic fatigue syndrome with reference to neurological and cognitive abnormalities. *Brain Research Bulletin* 55, 319–325.
- Verstraeten, S.V., Keen, C.L., Golub, M.S., Oteiza, P.I., 1998. Membrane composition can influence the rate of Al³⁺-mediated lipid oxidation: effect of galactolipids. *Biochemistry, J.* 333 (Pt 3), 833–888.
- Verstraeten, S.V., Oteiza, P.I., 2000. Effects of Al(3+) and related metals on membrane phase state and hydration: correlation with lipid oxidation. *Archives of Biochemistry and Biophysics* 375, 340–346.
- Wang, B.X. (Ed.), 1999. Contemporary Pharmacology of Chinese Herbal Medicine. Tianjin Science and Technology Press, Tianjin, pp. 1263–1264.
- Wills, M.R., Savory, J., 1983. Aluminium poisoning: dialysis encephalopathy, osteomalacia, and anaemia. *Lancet* 2, 29–34.
- Xie, Z.F., Lou, Z.Q., Huang, X.K. (Eds.), 1994. Classified Dictionary of Traditional Chinese Medicine. New World Press, Beijing, p. 385.
- Zatta, P., Nicosia, V., Zambenedetti, P., 1998. Evaluation of MAO activities on murine neuroblastoma cells upon acute or chronic treatment with aluminium (III) or tacrine. *Neurochemistry International* 32, 273–279.

- Zhang, Y.W., Xue, Z., 1991a. Studies on the chemical constituents of *Dipsacus asper* Wall. *Yao Xue Xue Bao* 26, 676–681.
- Zhang, Y.W., Xue, Z., 1991b. New triterpenoid glycosides from *Dipsacus asper* Wall. *Yao Xue Xue Bao* 26, 911–917.
- Zhang, Y.W., Xue, Z., 1992. Structure determination of saponin IX and X from *Dipsacus asper* Wall. *Yao Xue Xue Bao* 27, 912–917.
- Zhang, Y.W., Xue, Z., 1993. Structure determination of saponin XI, XII and XIII from *Dipsacus asper* Wall. *Yao Xue Xue Bao* 28, 358–363.
- Zhou, F.W., Shi, M.S., Zhou, J.L. (Eds.), 1981. *Chinese Herbology*. Shandong Science and Technology Press, Weifang, Shandong, pp. 596–597.